MILITARY SPECIFICATION
BELLOWS ASSEMBLY, FLEXIBLE, FOR LAUNCHING SYSTEM, MISSILE MARK 17 MODS 0, 1, 2, AND 3

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

SCOPE

1.1 Scope - This specification states the requirements for the flexible-bellows assembly used for bridging the intertube space between the outer tube and the launcher tube.

1.2 Classification - The flexible-bellows assembly shall be classified as follows:

Class 1 - Loose reinforcing rings (LD 512273)
Class 2 - Embedded reinforcing rings (Dwg 2024860)

APPLICABLE DOCUMENTS

2.1 Documents - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-F-21568	Fluids, Silicone, Dimethyl Polysiloxane

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS, Code Ident 10001

BuWep LD 512273	Flexible Bellows, SSB(N) Launcher
BuWep Dwg 2024860	Flexible Bellows

FSC 1440

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SA-2440165	Hydrotest Fixture for Flexible Bellows
SA-2440203	Life Testing Fixture for Upper and Lower Seals (Also used for bellows)
SA-2440379	Corrosion Test
SA-2440482	Bellows Submergence Test

PUBLICATIONS

NAVY DEPARTMENT

BUREAU OF NAVAL WEAPONS

OP 400	General Instruction for Design, Manufacture, and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Description - The bellows assembly is a flexible, corrugated neoprene tube with a static length of 9 inches, suitably restrained and reinforced by metal rings, and fastened to a metal mounting ring at each end.

3.2 General Requirements

3.2.1 Preproduction Sample - Unless otherwise specified in the contract or order, preproduction samples of the flexible-bellows assembly shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in Section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification. At the discretion of' the procuring agency, the requirement for the preproduction sample may be waived and shall then be satisfied by substitution of a production unit chosen at random by the Government Inspector.

3.2.2 Materials - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the flexible-bellows assembly shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BuWep LD 512273 or Dwg 2024860.

3.2.3 Finishes, Coatings, Construction, and Dimensions - Finishes, Coatings, construction, and dimensions for the flexible-bellows assembly shall be in accordance with the applicable drawings, specifications, and publications listed in BuWep LD 512273 or Dwg 2024860.

3.2.4 Marking - Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BuWep LD 512273 or Dwg 2024860.

3.2.5 Interchangeability - Individual assemblies and/or piece parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with like numbered parts.

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3.2.6 Gages - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, the Government will furnish drawings of pertinent Naval Final Inspection Gages for guidance in the design of the contractors inspection gages. However, such procedure shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Threads - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Requests for Deviation - Requests for deviation from this specification or any specifications, publications, materials, and processes specified herein, shall be as set forth in OP 400.

3.2.9 Conflicting Requirements - In the event of conflict between this specification and any specification, publication, or drawing referenced herein, this specification shall govern.

3.2.10 Tolerances - Unless otherwise specifically noted, tolerances and accuracies stated herein are absolute.

3.3 Performance Requirements and Product Characteristics

3.3.1 Hydrostatic Pressure - The flexible-bellows assembly shall withstand an internal hydrostatic pressure increased from 0 to 195-205 psig within 5 seconds, maintained for 10 minutes without leakage or damage, and thereafter shall meet all performance requirements and product characteristics of this specification. To satisfy this requirement, each flexible-bellows assembly shall be tested as specified in 4.5.4.2, and the preproduction sample shall be tested. as specified in 4.6.1.

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3.3.2 Operating Life and Flexure - The flexible-bellows assembly shall withstand 150 cycles of axial flexure and 150 cycles of lateral flexure at the rate of one cycle-per-minute. In axial flexure, the free end of the bellows is moved through an elliptical path having axes. of 8 and 12 inches, with the plane of motion and the minor axis perpendicular to the mounting ring. In lateral flexure, the free end of the bellows is moved through an elliptical path having axes of 8 and 12 inches, with the,Plane of motion parallel to and 9 inches from the mounting ring. flexing, the bellows shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.3.3 Deflection - The flexible-bellows assembly, in a horizontal position and with the ends fixed axially 9 inches apart, shall not deflect to the extent of obstructing the 4-1/2 inch diameter clear line of sight.

3.4 Environmental Conditions

3.4.1 Temperature and Humidity

3.4.1.1 Normal Temperature - The flexible-bellow assembly shall be capable of withstanding continuous sure to air temperature of +8o ± 3 degrees Fahrenheit with a relative humidity of 5O percent or less for 90 days, shall show no damage, and thereafter shall meet all performance requirements and product characteristics of this specification.

3.4.1.2 Low Temperature - The flexible-bellows assembly shall withstand temperature of +28 ± 4 degrees Fahrenheit for 4 hours, shall show no damage, and thereafter shall meet all performance requirements and product characteristics of this specification.

3.4.1.3 High Temperature - The flexible-bellows assembly shall withstand temperature of +110 ± 4 degrees Fahrenheit, with a relative humidity between 95 and 98 percent, for 12 hours, shall show no damage, and thereafter shall meet all performance requirements and product characteristics of this specification.

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3.4.1.4 Thermal Shock - The flexible-bellows assembly shall be capable of withstanding a temperature change from +80 ± 3 degrees to +30 ± 2 degrees Fahrenheit within 5 seconds, shall show no damage, and thereafter shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Submersion - The interior surface of the flexible-bellows assembly shall withstand submersion in sea water for 90 consecutive days at temperatures between +28 and +85 degrees Fahrenheit, shall show no damage, and thereafter shall meet all performance requirements and product characteristics of this specification.

3.4.3 Exposure to Hydraulic Fluid - The exterior of the flexible-bellows assembly shall withstand submersion in silicone fluid (Specification MIL-F-21568), at a temperature of +65 ± 5 degrees Fahrenheit, for 90 consecutive days, shall show no damage, and thereafter shall meet all performance requirements and product characteristics of this specification.

3.5 Cleanup - Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of all foreign matter. Burrs and sharp edges shall be removed.

3.6 Workmanship - The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's Inspection - Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.2 Government Inspection - The Government Inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings, and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400. Production tests of final assemblies shall be on a 100-percent basis. No sampling is permitted for acceptance inspection purposes. Only those assemblies which pass final inspection shall be accepted by the Navy; a rejected assembly may be reworked and submitted in accordance with OP 400. Failed parts and components shall be submitted to the Material Review Board for disposition.

4.2.1 Preproduction Sample - The preproduction sample specified in 3.2.1 shall be submitted to the Government Inspector for preproduction tests at an activity designated in the contract, or by the Bureau of Naval Weapons, for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the item by the contractor prior to approval by the contracting activity, or the completion of tests on the preproduction samples, shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.3 Classification of Tests

4.3.1 Production Tests - These tests are to be performed on each flexible-bellows assembly being submitted for acceptance under contract. Production tests shall be either performed by the manufacturer and witnessed by the Government Inspector or these tests may be conducted by the Government Inspector, These tests are detailed in 4.5.

4.3.2 Preproduction Tests - Preproduction tests are those which are performed on samples selected as specified in 4.2.1, which are representative of the production of the flexible-bellows assembly, to determine that the production meets the requirement of this specification. These are detailed in 4.5 and 4.6. Acceptance shall be based on no defects in the sample. Failure of the sample to comply with these requirements will result in the cessation of production until determination of cause of failure is made and appropriate correction approved by the procuring agency.

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4.3*3 Tests After Delivery - Unless otherwise specified in the contract, the Government reserves the right to conduct any or all of the tests specified herein as deemed necessary after installation of the flexible-bellows assembly. Any defects discovered before, during, or after these tests, due to the contractor's failure to comply with drawings or specifications for this equipment, shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.4 Special Test Equipment - The following items of special test equipment are required to perform the acceptance tests set forth in this specification:

SA-2440165	Hydrotest Fixture for Flexible Bellows
SA-2440203	Life Testing Fixture for Upper and Lower Seals (Also used for bellows)
SA-2440379	Corrosion Test
SA-2440482	Bellows Submergence Test

4.5 Production Tests

4.5.1 Responsibility - The contractor or Government Inspector shall perform production tests as specified in 4.3.1 to assure that the flexible-bellows assembly is in compliance with the requirements of 3.3.1 and 3.3.3.

4.5.2 Test Conditions - Unless otherwise specified, the flexible-bellows assembly shall be subjected to production tests under the following conditions:

Temperature:	Room ambient, +65 to +85 degrees Fahrenheit
Altitude:	Normal, ground atmospheric pressure
Vibration:	None
Humidity:	Room ambient

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4.5.3 Test and Inspection Equipment and Facilities - The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all production tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified shall be of sufficient accuracy and quality to permit performance of the required production tests.

4.5.4 Test Procedure

4.5.4.1 Materials and Dimensions - Each flexible-bellows assembly shall be inspected for conformance to the requirements Of 3.2.2.

4.5.4.2 Hydrostatic Pressure Test - Each flexible-bellows assembly, mounted in a test fixture which duplicates mating launcher tube and ship configuration, shall be subjected to an internal hydrostatic proof pressure of 195-205 psig for 10 minutes to meet the requirements of 3.3.1.

4.5.4.3 Deflection Test - each flexible-bellows assembly, mounted in a test fixture with the ends 9 inches apart and fixed axially in lines shall. be visually examined for conformance to the requirement of 3.3.3.

4.6 Preproduction Tests

4.6.1 Hydrostatic Pressure, Rapid Application Test - One flexible-bellows assembly, mounted in a test fixture which duplicates mating launcher tube and ship configuration, shall be subjected to an internal hydrostatic proof pressure increased from 0 to 195-205 psig within 5 seconds and maintained for 10 minutes to meet the requirements of 3.3.1.

4.6.2 Operating Life and Flexure Test - One flexible-bellows assembly, mounted in a suitable test fixture, shall be subjected to 150 cycles of axial flexure and 150 cycles of lateral flexure, at the rate of 1 cycle-per-minute, to meet the requirements of 3.3.2.

4.6.3 Submersion Test - To meet the requirement of 3.4.2, one internal surface of the flexible-bellows assembly shall be subjected to submersion in sea water for 90 consecutive days with a temperature of 65 ± 20 degrees Fahrenheit for 88 days and with temperatures between +28 and +32 degrees Fahrenheit for 2 days.

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4.6.4 Low Temperature Test - One flexible-bellows assembly shall be subjected to a temperature of +28 ± 4 degrees Fahrenheit for 4 hours, to meet the requirements of 3.4.1.2.

4.6.5 High Temperature Test -One flexible-bellows assembly shall be subjected to a temperature of +110 ± 4 degrees Fahrenheit, with a relative humidity between 95 and 98 percent, for 12 hours, to meet the requirements of 3.4.1.3.

4.6.6 Thermal Shock Test - One flexible-bellows assembly, mounted in a suitable test fixture, shall be subjected to a temperature change from +80 ± 3 degrees to +30 ± 2 degrees Fahrenheit to meet the requirements of 3.4.1.4.

4.6.7 Hydraulic Fluid Test - The external surface of one flexible-bellows assembly, shall be submerged in silicone fluid (Specification MIL-F-21568) at a temperature of +65 ± 5 degrees Fahrenheit, for 90 consecutive days, to meet the requirements of 3.4.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Preservation and packaging of the flexible-bellows assembly shall be Level A or Level G.

5.1.1 Level A - Level A shall be as specified by the procuring activity.

5.1.2 Level C. - Cleaning, drying, preservation, and unit packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing of the flexible-bellows assembly shall be Level A or Level C.

5.2.1 Level A - Level A shall be as specified by the procuring activity.

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5.2.2 Level C - The flexible-bellows assembly, packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with Uniform Freight Classification Rules and other common-carrier regulations applicable to the mode of transportation.

5.3 Marking - In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standards MIL-STD-129.

6. NOTES

6.1 Ordering Data - Procurement documents should specify the following:

(a) Title, numbers and date of this specification.

(b) Selection of applicable levels of preservation, packaging, and packing.

NOTICE : When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture use, or sell any patented invention that may in any way be related thereto,

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